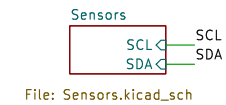
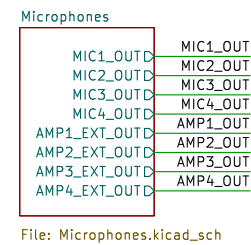
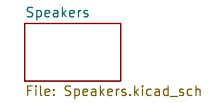
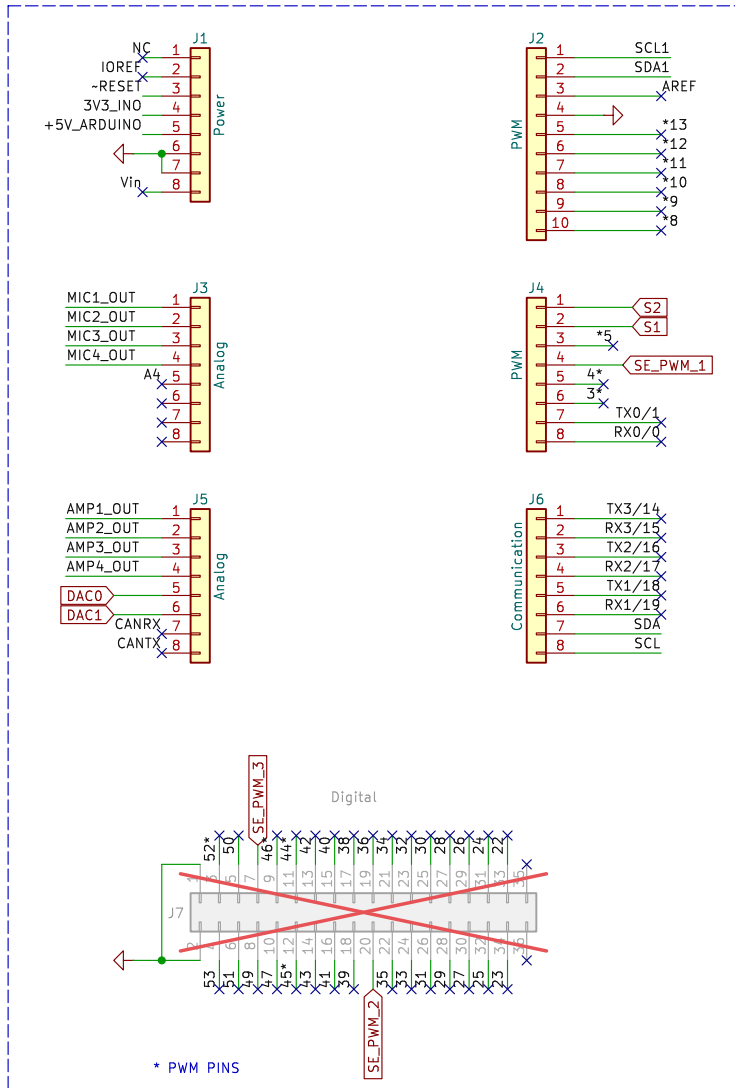


ARDUINO CONNECTORS



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Infineon Technologies

Sheet: /
File: SensEdu.kicad_sch

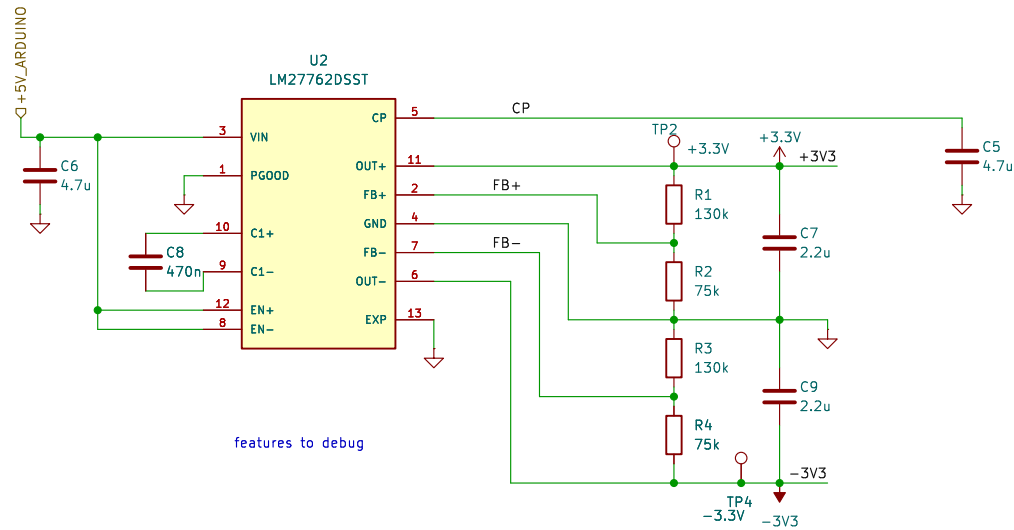
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KiCad E.D.A. 9.0.2

Date: 2026-02-26

Rev: 0.9
Id: 1/6

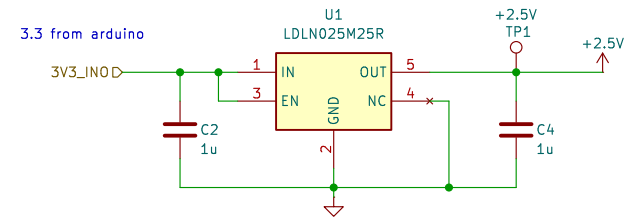
CHARGE PUMP + LDO FOR GENERATING $\pm 3.3V$



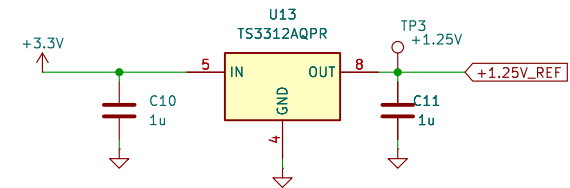
$$V_{out-} = -1.2V \cdot (R_{39} + R_{40}) / R_{40}; R_{40} > 50k\Omega; R_{40} = 75k\Omega; R_{39} = 130k\Omega \text{ OR } R_{40} = 91k\Omega; R_{39} = 160k\Omega$$

$$V_{out+} = 1.2V \cdot (R_{41} + R_{42}) / R_{42}; R_{42} > 50k\Omega; R_{42} = 75k\Omega; R_{41} = 130k\Omega \text{ OR } R_{42} = 91k\Omega; R_{41} = 160k\Omega$$

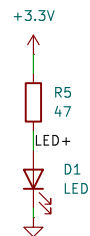
LOW DROPOUT VOLTAGE REFERENCE 3.3V $\rightarrow$ 2.5V



LOW DROPOUT VOLTAGE REFERENCE 3.3V $\rightarrow$ 1.25V



POWER INDICATION



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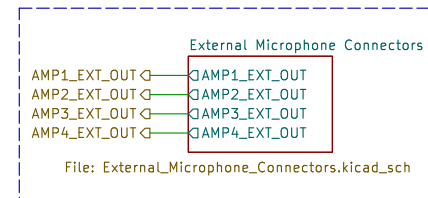
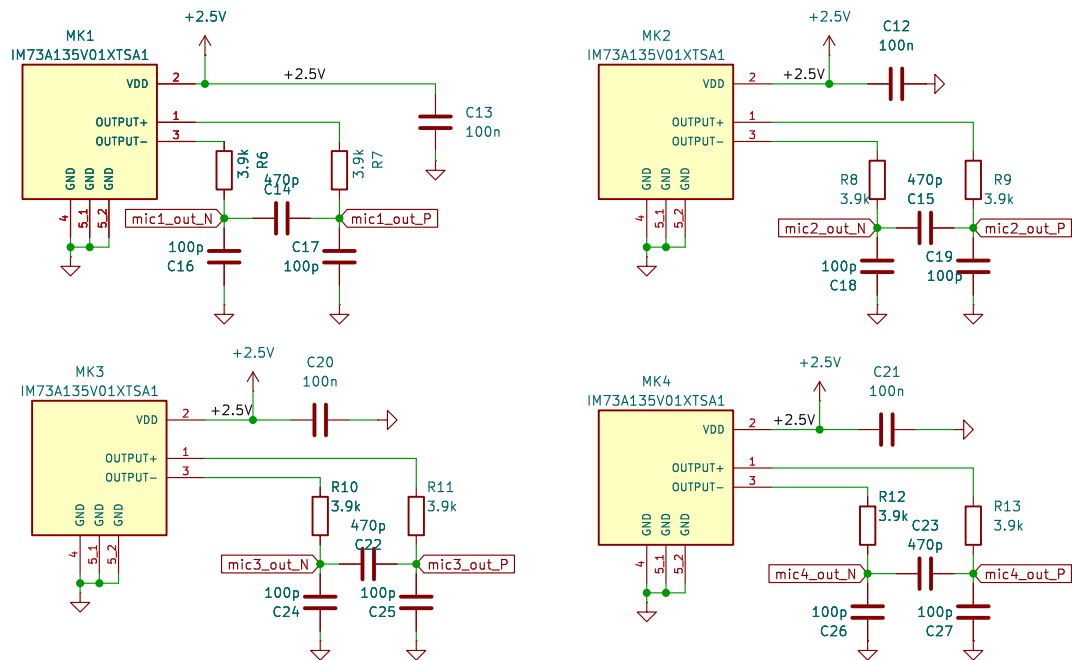
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Title: SensEdu

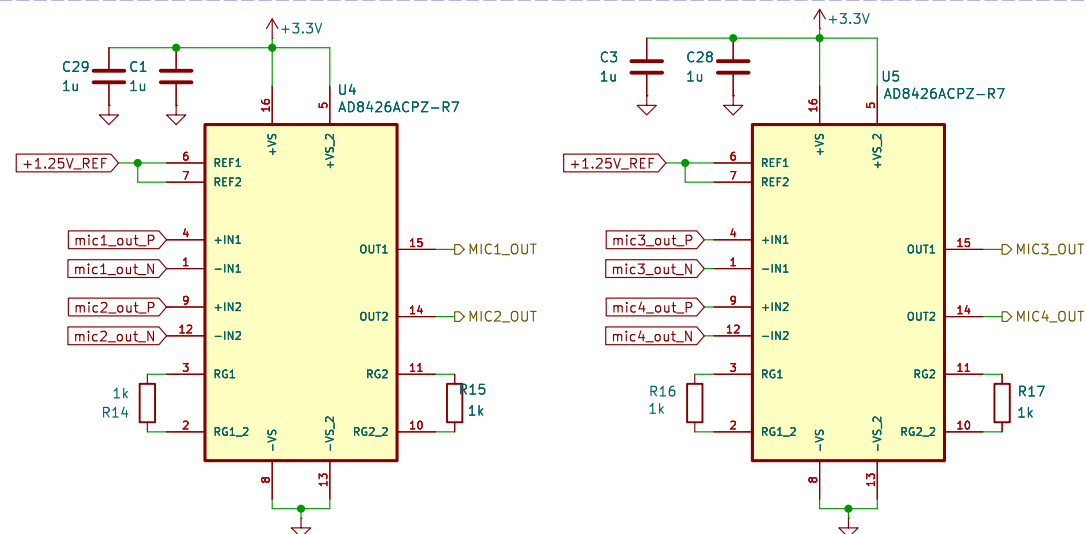
Size: A4 Date: 2026-02-26
 KiCad E.D.A. 9.0.2

Rev: 0.9
 Id: 2/6

MEMS MICROPHONES



MICROPHONE AMPLIFIERS



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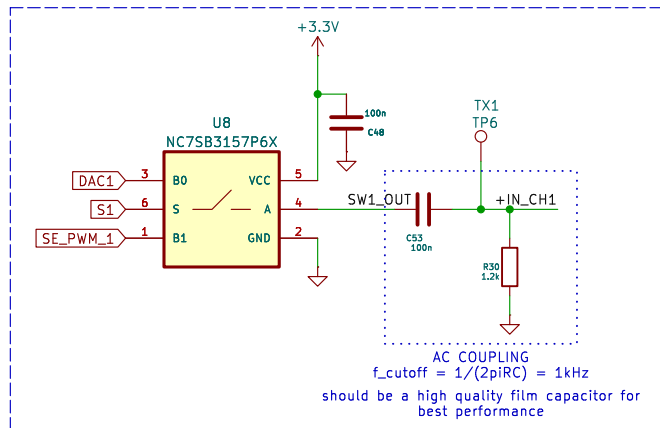
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Title: SensEdu

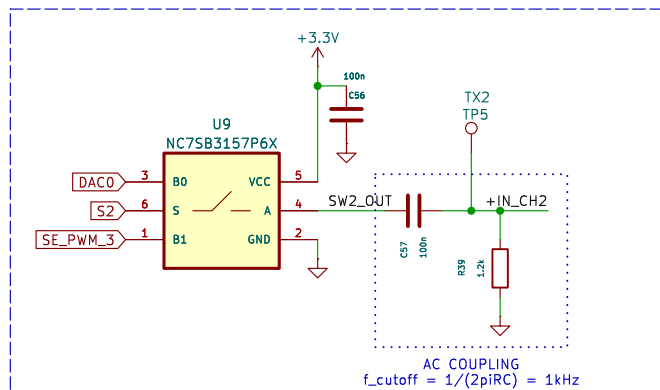
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KiCad E.D.A. 9.0.2

Rev: 0.9
Id: 3/6

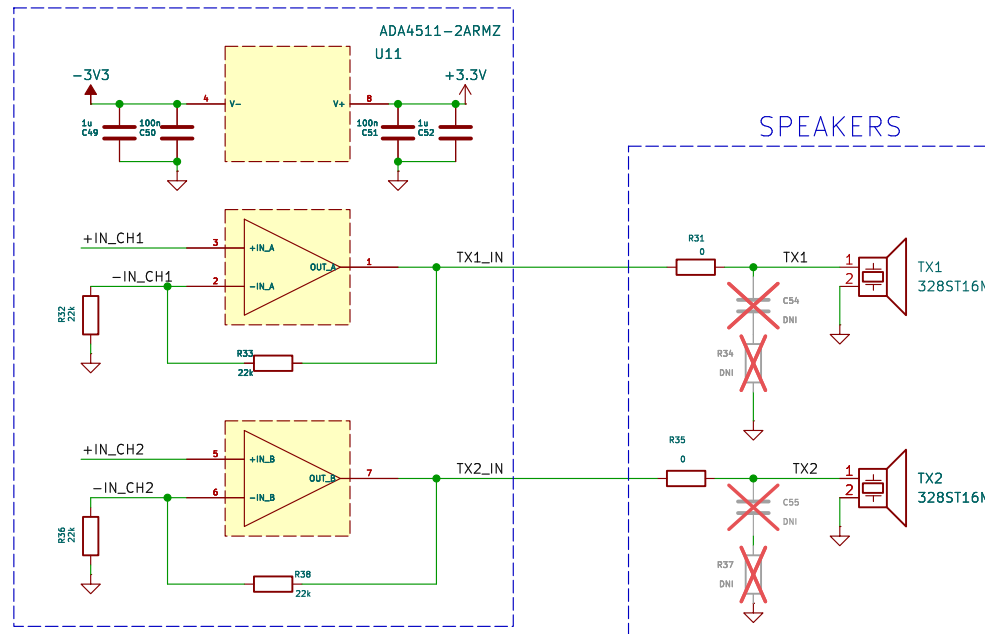
SWITCH WITH FILTER FOR DIFFERENT DRIVING SOURCES



SWITCH WITH FILTER FOR DIFFERENT DRIVING SOURCES



DUAL CHANNEL PREAMPLIFIER



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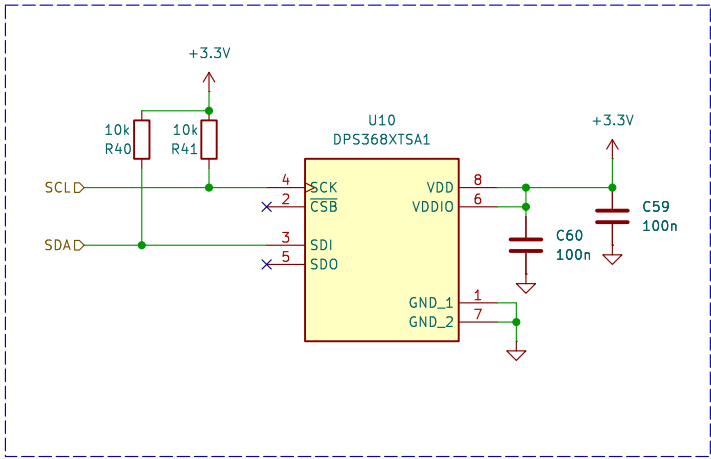
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Title: SensEdu

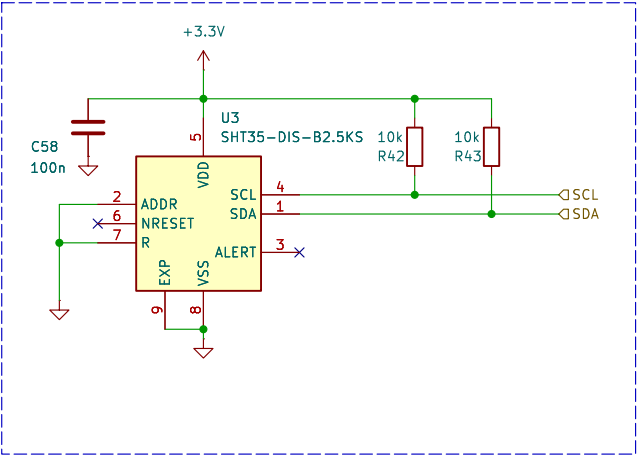
Size: A4 Date: 2026-02-26
 KiCad E.D.A. 9.0.2

Rev: 0.9
 Id: 5/6

PRESSURE SENSOR



HUMIDITY SENSOR



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Sheet: /Sensors/
File: Sensors.kicad_sch

Title: SensEdu

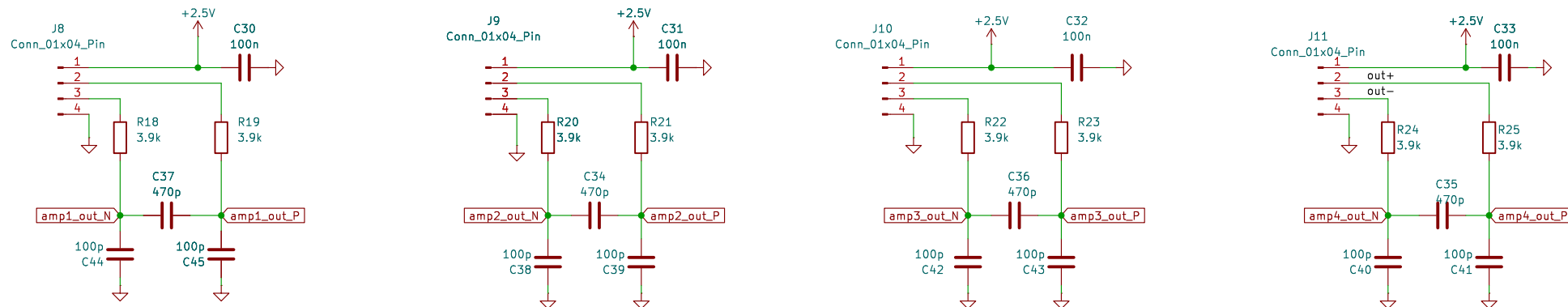
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KiCad E.D.A. 9.0.2

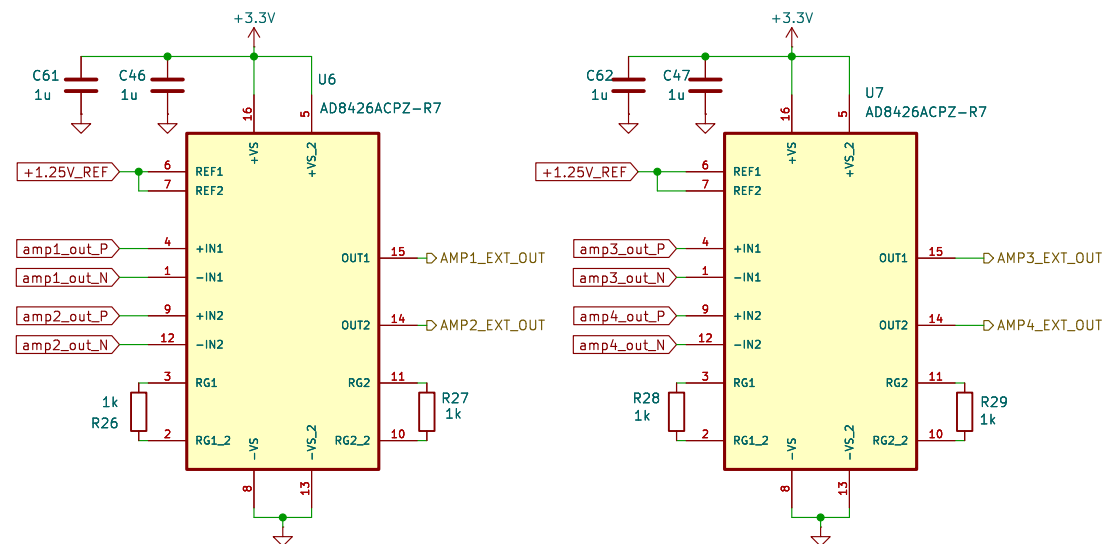
Rev: 0.9

Id: 6/6

EXTERNAL MICROPHONE CONNECTORS



MICROPHONE AMPLIFIERS



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Sheet: /Microphones/External Microphone Connectors/
 File: External_Microphone_Connectors.kicad_sch

Title: SensEdu

Size: A4 Date: 2026-02-26
 KiCad E.D.A. 9.0.2

Rev: 0.9
 Id: 4/6